

CariSurg MedTech Pathways

AI-Harm Case Study

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Student GitHub: <https://github.com/imaniiz/CariSurg-Portfolio/tree/main>

Overview: This document provides evidence of completion of the Real-world AI harm case study

Real-World Harm Case Study

Artificial intelligence has the potential to improve clinical decision-making by analysing large amounts of patient data and supporting healthcare professionals. However, AI systems can also unintentionally reinforce existing biases if they are developed or implemented without appropriate safeguards. A documented example is the use of algorithms to prioritise patients for organ transplants. As discussed by Denecke et al. (2025)(1), the United Network for Organ Sharing (UNOS) algorithm in the United States was found to unfairly delay kidney transplants for Black patients by overestimating kidney function, resulting in longer waiting times for life-saving treatment. Similar concerns were identified with the United Kingdom's Transplant Benefit Score algorithm, which assigned lower priority scores to younger patients awaiting liver transplants. The primary cause of this incident was algorithmic bias embedded within the decision-making process. Rather than making recommendations based solely on clinical need, the algorithms incorporated variables and assumptions that produced unequal outcomes for different patient groups. In the UNOS case, the algorithm relied on an estimated kidney function equation that included race as a variable. This caused kidney disease severity in many Black patients to be overestimated, making them appear healthier than they actually were and delaying their eligibility for transplantation. In the UK example, the prioritisation algorithm disadvantaged younger patients by assigning them lower transplant benefit scores. These cases demonstrate how biased assumptions within an algorithm can directly influence access to healthcare.

Another contributing factor was insufficient evaluation of fairness before deployment. Although these algorithms were designed to improve consistency and efficiency, there was limited monitoring to determine whether recommendations differed systematically across demographic groups. Without regular auditing, discriminatory outcomes remained undetected until patients experienced significant delays in receiving treatment. The consequences of these failures extended beyond technical inaccuracies. Delayed organ transplantation can lead to worsening health outcomes, reduced quality of life and, in severe cases, increased mortality risk. The paper notes that one affected patient waited more than five years for a

kidney transplant because of the algorithm's bias. Such incidents also undermine public trust in AI-assisted healthcare and raise ethical concerns regarding fairness, transparency and equitable access to medical treatment. Several measures could reduce the likelihood of similar incidents occurring in future AI systems. Algorithms should undergo comprehensive fairness testing using diverse and representative patient populations before clinical deployment. Healthcare organisations should regularly audit AI systems to identify differences in performance across demographic groups and update models when disparities are detected. Most importantly, AI should function as a clinical decision-support tool rather than replacing healthcare professionals. Human oversight allows clinicians to recognise recommendations that appear inconsistent with a patient's actual condition and intervene when necessary. This case highlights that AI systems are only as fair as the data, assumptions and evaluation methods used to develop them. For AI-assisted emergency department triage systems, ensuring representative datasets, continuous monitoring and meaningful human oversight will be essential to prevent similar biases from affecting patient care and to support safe, equitable clinical decision-making.

References

1. Denecke K, Lopez-Campos G, Rivera-Romero O, Gabarron E. The Unexpected Harms of Artificial Intelligence in Healthcare: Reflections on Four Real-World Cases. In: Bürkle T, Afzali M, Denecke K, Kim SI, Krummrey G, Thilo FJS, et al., editors. *Studies in Health Technology and Informatics* [Internet]. IOS Press; 2025 [cited 2026 Jun 27]. Available from: <https://ebooks.iospress.nl/doi/10.3233/SHTI250219> doi:10.3233/SHTI250219